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High precision sensor CAN protocol

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Instructions for use：

The serial port sending command must be completed within 10S, otherwise it will be automatically locked. In order to avoid automatic locking, the following steps can be performed first.

1. Enter the unlock command

2. Enter the command that needs to modify or read the data

3. Save the command

Register table

ADDR (Hex)	ADDR (Dec)	REGISTER NAME	FUNCTION	SERIAL I/F	Bit15
00	00	SAVE	save/reboot/reset	R/W	SAVE[15:0]
01	01	CALSW	calibration mode	R/W	
04	04	BAUD	serial baud rate	R/W	
05	05	AXOFFSET	Acceleration X Zero Bias	R/W	AXOFFSET [15:0]
06	06	AYOFFSET	Acceleration Y zero Bias	R/W	AYOFFSET [15:0]
07	07	AZOFFSET	Acceleration Z zero Bias	R/W	AZOFFSET [15:0]
08	08	GXOFFSET	Angular velocity X zero bias	R/W	GXOFFSET [15:0]
09	09	GYOFFSET	Angular velocity Y zero bias	R/W	GYOFFSET [15:0]
0A	10	GZOFFSET	Angular velocity Z zero bias	R/W	GZOFFSET [15:0]
0B	11	HXOFFSET	Magnetic Field X Zero Bias	R/W	HXOFFSET [15:0]

0C	12	HYOFFSET	Magnetic Field Y zero bias	R/W	HYOFFSET [15:0]
0D	13	HZOFFSET	Magnetic Field Z zero bias	R/W	HZOFFSET [15:0]
0E	14	WORKMODE【1】	Z-axis operation mode	R/W	WORKMODE[15:0]
10	16	GYROPTP【1】	Z-axis static peak-to-peak	R/W	GYROPTP[15:0]
11	17	GPTPTIME【1】	Z-axis peak-to-peak acquisition time	R/W	GPTPTIME [15:0]
12	18	GYROBAIS【1】	Z-axis zero bias value	R/W	GYROBAIS [15:0]
13	19	GBAISTIME【1】	Z-axis zero bias acquisition time	R/W	GBAISTIME[15:0]
14	20	GSTATICTHRE【1】	Z-axis static threshold	R/W	GSTATICTHRE[15:0]
15	21	GSTATICTIME【1】	Z axis stabilization time	R/W	GSTATICTIME[15:0]
16	22	PGSCALE【1】	Z-axis calibration factor P	R/W	PGSCALE[15:0]
18	24	GSCALERANGE【1】	Z-axis calibration angle	R/W	GSCALERANGE[15:0]

1A	26	IICADDR	Device address	R/W	
1B	27	LEDOFF	Turn off the LED lights	R/W	
1C	28	MAGRANG X	Magnetic Field X Calibration Range	R/W	MAGRANG X[15:0]
1D	29	MAGRANG Y	Magnetic Field Y Calibration Range	R/W	MAGRANG Y[15:0]
1E	30	MAGRANG Z	Magnetic Field Z Calibration Range	R/W	MAGRANG Z[15:0]
1F	31	BANDWID TH	Bandwidth	R/W	
20	32	GYRORAN GE	Gyroscope range	R/W	
21	33	ACCRANG E	Acceleratio n range	R/W	
22	34	SLEEP	sleep	R/W	
23	35	ORIENT	Installation direction	R/W	
24	36	AXIS6	algorithm	R/W	
25	37	FILTK	Dynamic filtering	R/W	FILTK[15:0]
26	38	GPSBAUD	GPS baud rate	R/W	
27	39	READADD R	read register	R/W	

2A	42	ACCFILT	acceleration filter	R/W	ACCFILT[15:0]
2E	46	VERSION	version number	R	VERSION[15:0]
30	48	YYMM	year/mouth	R/W	MOUTH[15:8]
31	49	DDHH	day/hour	R/W	HOUR[15:8]
32	50	MMSS	minute/second	R/W	SECONDS[15:8]
33	51	MS	millisecond	R/W	MS[15:0]
34	52	AX	Acceleration X	R	AX[15:0]
35	53	AY	Acceleration Y	R	AY[15:0]
36	54	AZ	Acceleration Z	R	AZ[15:0]
37	55	GX	Angular velocity X	R	GX[15:0]
38	56	GY	Angular velocity Y	R	GY[15:0]
39	57	GZ	Angular velocity Z	R	GZ[15:0]
3A	58	HX	Magnetic field X	R	HX[15:0]
3B	59	HY	Magnetic field Y	R	HY[15:0]
3C	60	HZ	Magnetic field Z	R	HZ[15:0]
3D	61	LRoll	X-axis angle low	R	LRoll[15:0]

3E	62	HRoll	X-axis angle high	R	HRoll[15:0]
3F	63	LPitch	Y-axis angle low	R	LPitch[15:0]
40	64	HPitch	Y-axis angle high	R	HPitch[15:0]
41	65	LYaw	Z-axis angle low	R	LYaw[15:0]
42	66	HYaw	Z-axis angle high	R	HYaw[15:0]
43	67	TEMP	Module temperature	R	TEMP[15:0]
61	97	GYROCALITHR	Gyro Calibration Threshold	R/W	GYROCALITHR[15:0]
63	99	GYROCALTIME	Gyro calibration time	R/W	GYROCALTIME[15:0]
69	105	KEY	unlock	R/W	KEY[15:0]
6A	105	WERROR	Gyroscope change value	R/W	WERROR[15:0]
6E	110	WZTIME	Z-axis still time threshold	R/W	WZTIME[15:0]
6F	111	WZSTATIC	Z-axis static threshold	R/W	WZSTATIC[15:0]
74	116	MODDELAY	485 data response delay	R/W	MODDELAY[15:0]
7F	127	NUMBERID1	Device No. 1-2	R	ID2[15:8]

80	128	NUMBERID 2	Device No. 3-4	R	ID4[15:8]
81	129	NUMBERID 3	Device No. 5-6	R	ID6[15:8]
82	130	NUMBERID 4	Device No. 7-8	R	ID8[15:8]
83	131	NUMBERID 5	Device No. 9-10	R	ID10[15:8]
84	132	NUMBERID 6	Device No. 11-12	R	ID12[15:8]
95	149	LREFROLL	ROLL angle zero is low bit	R/W	LREFROLL[15:0]
96	150	HREFROLL	ROLL angle zero bias high bit	R/W	HREFROLL[15:0]
97	151	LREFPITCH	PITCH angle zero bias low bit	R/W	LREFPITCH[15:0]
98	152	HREFPITCH	PITCH angle zero bias high bit	R/W	HREFPITCH[15:0]

explanation：

Callout registers are only available for HWTX073.

Protocol format

Active output format

- Data is sent in hexadecimal, not ASCII.
- Each data is transmitted in sequence by low byte and high byte, and the two are combined into a signed short type of data. For example, for data DATA1, DATA1L is the low byte and DATA1H is the high byte. The conversion method is as follows: Suppose DATA1 is the actual data, DATA1H is its high-byte part, DATA1L is its low-byte part,Then: DATA1=(short)((short)DATA1H<<8|DATA1L). It must be noted here that DATA1H needs to be coerced into a signed short type of data before shifting, and the data type of DATA1 is also a signed short type, so that it can represent negative numbers.

protocol head	Data content	Data lower 8 bits	Data high 8 bits	Data lower 8 bits	Data high 8 bits
0x55	TYPE【1】	DATA1L[7:0]	DATA1H[15:8]	DATA2L[7:0]	DATA2H[15:8]

【1】TYPE(Data content):

TYPE	Remark
0x50	Time
0x51	Acceleration
0x52	Angular velocity
0x53	Angle
0x54	Magnetic field
0x5F	Read

Time output

0x55	0x50	YY	MM	DD	HH
Name	Description	Remark			
YY	year				
MM	mouth				
DD	day				
HH	hour				
MN	minute				
SS	second				

Acceleration output

0x55	0x51	AxL	AxH	AyL	AyH
Name	Description	Remark			
AxL	Acceleration X low 8 bits	acceleration X=((AxH<<8)|AxL)/32768*16g (g is the gravitational acceleration, preferably 9.8m/s2)			
AxH	Acceleration X high 8 bits				
AyL	Acceleration Y low 8 bits	acceleration Y=((AyH<<8)|AyL)/32768*16g (g is the gravitational acceleration, preferably 9.8m/s2)			
AyH	Acceleration Y high 8 bits				

AzL	Acceleration Z low 8 bits	acceleration Z=((AzH<<8)|AzL)/32768*16g (g is the gravitational acceleration, preferably 9.8m/s2)			
AzH	Acceleration Z high 8 bits				

Angular velocity

0x55	0x52	WxL	WxH	WyL	WyH
Name	Description	Remark			
WxL	Angular velocity X low 8 bits	Angular velocity X=((WxH<<8)|WxL)/32768*2000°/s			
WxH	Angular velocity X high 8 bits				
WyL	Angular velocity Y low 8 bits	Angular velocity Y=((WyH<<8)|WyL)/32768*2000°/s			
WyH	Angular velocity Y high 8 bits				
WzL	Angular velocity Z low 8 bits	Angular velocity Z=((WzH<<8)|WzL)/32768*2000°/s			
WzH	Angular velocity Z high 8 bits				

Angle output(Roll)

0x55	0x53	0x01	0x00	LRollL	LRollH
Name	Description	Remark			
0x01	Roll angle X mark	Indicates that this packet is roll angle data			
0x00	Retain	Retain			
LRollL	roll angle X[8:0]	roll angle （x-axis） Roll = (float)(((int)HRollH<<24)|((int)HRollL<<16)| ((int)LRollH<<8)|(int)LRollL)/1000.0			
LRollH	roll angle X[15:9]				
HRollL	roll angle X[23:16]				
HRollH	roll angle X[31:24]				

Angle output (Pitch)

0x55	0x53	0x02	0x00	LPtichL	LPtichH
Name	Descriptio n	Remark			
0x01	Pitch angle Y mark	Indicates that this packet is pitch angle data			
0x00	retain	retain			
LRollL	Pitch Y[8:0]	Pitch angle （y-asix）			
LRollH	Pitch Y[15:9]				
HRollL	Pitch Y[23:16]				
HRollH	Pitch Y[31:24]				

Angle output(Yaw)

0x55	0x53	0x03	0x00	LYawH	LYawH
Name	Description	Remark			
0x01	Yaw angle Z mark	Indicates that this data packet is yaw angle data			
0x00	retain	retain			
LRollL	yaw angleZ[8:0]	yaw angle（z-axis)			
LRollH	yaw angleZ[15:9]				
HRollL	yaw angleZ[23:16]				
HRollH	yaw angleZ[31:24]				

Magnetic field

0x55	0x54	HxL	HxH	HyL	HyH
Name	Description	Remark			
HxL	Magnetic field X lower 8 bits	Magnetic field X=((HxH<<8)|HxL)			
HxH	Magnetic field X high 8 bits				
HyL	Magnetic field Y lower 8 bits	Magnetic fieldY=((HyH <<8)|HyL)			
HyH	Magnetic field Y high 8 bits				
HzL	Magnetic field Z lower 8 bits	Magnetic fieldZ=((HzH<<8)|HzL)			
HzH	Magnetic field Z high 8 bits				

Read register format

- Data is sent in hexadecimal, not ASCII.

- Each register address, the number of read registers, and the read data are represented by two bytes. The high and low bits of the register address are represented by ADDRH and ADDRL, the high and low bits of the number of registers to be read are represented by LENH and LENL, and the high and low bits of the read data are represented by DATA1H and DATA1L.

command send

protocol header	protocol header	register	data lower 8 bits	Data high 8 bits
0xFF	0xAA	ADDR	DATAL[7:0]	DATAH[15:8]

Data return

0x55	0x5F	REG1L	REG1H	REG2L	REG2H
Name	Description	Remark			
REG1L	Register 1 lower 8 bits	REG1[15:0] = ((REG1H<<8)|REG1L)			
REG1H	Register 1 high 8 bits				
REG2L	Register 2 lower 8 bits	REG2[15:0] = ((REG2H<<8)|REG2L)			
REG2H	Register 2 high 8 bits				
REG3L	Register 3 lower 8 bits	REG3[15:0] = ((REG3H<<8)|REG3L)			
REG3H	Register 3 high 8 bits				

Write register format

- Data is sent in hexadecimal, not ASCII.
- All settings need to operate the unlock register (KEY) first
- For each register address, the write data is represented by two bytes. The high and low bits of the register address are represented by ADDRH and ADDRL, and the high and low bits of the written data are represented by DATAH and DATAL.

protocol header	protocol header	register	data lower 8 bits	data high 8 bits
0xFF	0xAA	ADDR	DATAL[7:0]	DATAH[15:8]

Register Description

SAVE（save/reboot/reset）

Register Name: SAVE Register Address: 0 (0x00) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	SAVE[15:0]	save: 0x0000 reboot: 0x00FF reset: 0x0001
example：FF AA 00 FF 00（reboot）		

CALSW（calibration mode）

Register Name: CALSW Register Address: 1 (0x01) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	CAL[3:0]	Set calibration mode: 0000(0x00): normal working mode 0001(0x01): Auto calibration 0011(0x03): height reset 0100(0x04): Set the heading angle to zero 0111(0x07): Magnetic Field Calibration (Spherical Fitting) 1000(0x08): set angle reference 1001(0x09): Magnetic Field Calibration (Dual Plane Mode)
example：FF AA 01 04 00（Heading angle set to zero）		

RSW（output content）

Register Name: RSW Register Address: 2 (0x02) Read and write direction: R/W Default: 0x001E		
Bit	NAME	FUNCTION
15:11		
10	GSA (0x5A)	0: turn off 1: turn on
9	QUATER (0x59)	0: turn off 1: turn on
8	VELOCITY (0x58)	0: turn off 1: turn on
7	GPS (0x57)	0: turn off 1: turn on
6	PRESS (0x56)	0: turn off 1: turn on
5	PORT (0x55)	0: turn off 1: turn on
4	MAG (0x54)	0: turn off 1: turn on
3	ANGLE (0x53)	0: turn off 1: turn on
2	GYRO (0x52)	0: turn off 1: turn on
1	ACC (0x51)	0: turn off 1: turn on
0	TIME (0x50)	0: turn off 1: turn on
example：FF AA 02 3E 00（Set to output only acceleration, angular velocity, angle, magnetic field, port status）		

RRATE（output rate）

Register Name: RRATE Register Address: 3 (0x03) Read and write direction: R/W Default: 0x0006		
Bit	NAME	FUNCTION
15:4		
3:0	RRATE[3:0]	Set the output rate: 0001(0x01): 0.2Hz 0010(0x02): 0.5Hz 0011(0x03): 1Hz 0100(0x04): 2Hz 0101(0x05): 5Hz 0110(0x06): 10Hz 0111(0x07): 20Hz 1000(0x08): 50Hz 1001(0x09): 100Hz 1011(0x0B): 200Hz 1011(0x0C): single postback 1100(0x0D): don't post back
Example: FF AA 03 03 00 (set 1Hz output)		

BAUD（Serial port baud rate）

Register Name: BAUD Register Address: 4 (0x04) Read and write direction: R/W Default: 0x0002		
Bit	NAME	FUNCTION
15:4		
3:0	BAUD[3:0]	Set the serial port baud rate: 0000(0x00): 1000000bps 0001(0x01): 800000bps 0010(0x02): 500000bps 0011(0x03): 400000bps 0100(0x04): 250000bps 0101(0x05): 200000bps 0110(0x06): 125000bps 0111(0x07): 100000bps 1000(0x08): 80000bps 1001(0x09): 50000bps 1010(0x0A): 40000bps 1011(0x0B): 20000bps 1100(0x0C): 10000bps 1101(0x0D): 5000bps 1110(0x0E): 3000bps
Example: FF AA 04 02 00 (set serial port baud rate 500000)		

AXOFFSET~HZOFFSET（Zero bias setting）

Register Name: AXOFFSET~HZOFFSET Register Address: 5~13 (0x05~0x0D) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	AXOFFSET[15:0]	Acceleration X-axis bias, actual acceleration bias=AXOFFSET[15:0]/10000(g)
15:0	AYOFFSET[15:0]	Acceleration Y-axis bias, actual acceleration bias=AYOFFSET[15:0]/10000(g)
15:0	AZOFFSET[15:0]	Acceleration Z-axis bias, actual acceleration bias=AZOFFSET[15:0]/10000(g)
15:0	GXOFFSET[15:0]	Angular velocity X-axis bias, actual angular velocity bias=GXOFFSET[15:0]/10000(°/s)
15:0	GYOFFSET[15:0]	Angular velocity Y-axis offset, actual angular velocity offset=GYOFFSET[15:0]/10000(°/s)
		Angular velocity Z-axis offset,

15:0	GZOFFSET[15:0]	actual angular velocity offset=GZOFFSET[15:0]/10000(°/s)
15:0	HXOFFSET[15:0]	Magnetic field X-axis zero bias
15:0	HYOFFSET[15:0]	Magnetic field Y axis zero bias
15:0	HZOFFSET[15:0]	Magnetic field Z axis zero bias
Example：FF AA 05 E8 03（Set acceleration X-axis zero bias 0.1g）,0x03E8=1000，1000/10000=0.1(g)		

WORKMODE（Z-axis operation mode）

Register Name: WORKMODE Register address: 14(0x0E) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	WORKMODE[3:0]	Set the Z-axis operation mode: 0000(0x00): normal data mode 0001(0x01): peak-to-peak mode 0010(0x02): seek zero offset mode 0011 (0x03): find scale factor mode
Example: Send: FF AA 14 01 00 (Automatically obtain zero offset)		

GYROPTP（Z-axis static peak-to-peak）

Register Name: GYROPTP Register address: 16(0x10) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	GYROPTP[15:0]	Parameters used in calculating zero offset, no need to set for sensor automatic acquisition
Write 0x01 to the "WORKMODE" register to enter the "peak-to-peak mode", in this mode the sensor automatically calculates the maximum and minimum values of the Z-axis angular velocity within the time set by "GPTPTIME" and records them in "GYROPTP". In "Zero Offset Mode", this data will be used to calculate and filter the zero offset. Z-axis static peak-to-peak value = GYROPTP/1000 (°/s) Example: Send: FF AA 27 10 00 (read Z-axis static peak-to-peak value) Return: 55 5F 64 00 XX XX XX XX XX XX SUM		

0x0064=100, Z-axis static peak-to-peak value = 100/1000=0.1 (°/s)		

GPTPTIME（Z-axis peak-to-peak acquisition time）

Register Name: GPTPTIME Register address: 17(0x11) Read and write direction: R/W Default: 0x000A		
Bit	NAME	FUNCTION
15:0	GPTPTIME[15:0]	Calculate the peak-to-peak time, the default is 10S
Example: Send: FF AA 11 0A 00 (set the Z-axis peak-to-peak acquisition time to 10S) After entering the "Peak-to-Peak Mode", obtain the difference between the maximum value and the minimum value of the Z-axis angular velocity within the "GPTPTIME" time, store it in "GYROPTP", and use this data for zero bias in the "Zero Bias Mode". calculation filter.		

GYROBAIS（Z axis zero bias value）

Register Name: GYROBAIS Register address: 18(0x12) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	GYROBAIS[15:0]	The zero bias value of the horizontal gyroscope, which can be obtained through the "Zero Bias Mode"
The Z-axis horizontal gyroscope has a certain zero bias value under static placement, and the angular velocity at rest can be eliminated by this zero bias value. The zero offset value can be automatically calculated by automatically obtaining the zero bias. Enter the "Zero bias Mode" to automatically calculate the zero bias value according to the peak-to-peak value of "GYROPTP" and the "GBAISTIME" zero bias acquisition time. Z-axis zero bias =GYROBAIS/1000（°/s） Example：		

Send：FF AA 27 12 00 （Read Z-axis static peak-to-peak value） Return：55 5F 64 00 XX XX XX XX XX XX SUM 0x0064=100，Z-axis static peak-to-peak= 100/1000=0.1（°/s）		

GBAISTIME（Z axis zero bias acquisition time）

Register Name: GBAISTIME Register Address: 19(0x13) Read and write direction: R/W Default: 0x000A		
Bit	NAME	FUNCTION
15:0	GBAISTIME[15:0]	Time required to calculate the zero bias value of the horizontal gyroscope
Example： Send：FF AA 13 0A 00 （Set the Z axis zero bias acquisition time to 10S） The time required to obtain the Z-axis zero bias, and the zero bias value is obtained according to this time when the zero bias is obtained。		

GSTATICTHRE（Z-axis static threshold）

Register Name: GSTATICTHRE Register address: 20(0x14) Read and write direction: R/W Default: 0x0032		
Bit	NAME	FUNCTION
15:0	GSTATICTHRE[15:0]	The larger the GSTATICTHRE[15:0], the weaker the seismic performance and the smaller the error The smaller the GSTATICTHRE[15:0], the better the seismic performance and the larger the error. Default value 50

The Z-axis horizontal gyroscope has slight data jitter when it is placed at rest. This parameter can filter out these slight jitters. When the angular velocity is less than the "GSTATICTHRE" setting value and lasts for the time set by "GSTATICTIME", it is regarded as stationary, and the Z-axis angular velocity is zero. This parameter can be appropriately increased when it is used in scenes with jitter and the Z-axis is accumulated due to jitter, and this parameter can be appropriately decreased in scenes with very slow and uniform rotation. Z-axis zero bias value=GSTATICTHRE/1000（°/s） Example： Send：FF AA 14 64 00 （(Set the Z-axis static threshold to 0.1g）,0x0064=100，100/10000=0.1(°))		

GSTATICTIME（Z axis stabilization time）

Register Name: GSTATICTIME Register address: 21(0x15) Read and write direction: R/W Default: 0x0064		
Bit	NAME	FUNCTION
15:0	GSTATICTIME[15:0]	Z-axis judges the still time
Z-axis static judgment time threshold. It is considered stationary when the angular velocity is less than the value set by "GSTATICTHRE" for the time set by "GSTATICTIME". If the stabilization time is required to be higher, this parameter can be adjusted appropriately. Lowering this parameter can speed up settling time but may also increase error. Z-axis stabilization time = GSTATICTIME/1000 (s) Example： Send：FF AA 15 32 00（(Set the Z axis stabilization time0.05s）,0x0032=50，50/10000=0.05(°))		

PGSCALE（Z-axis calibration factor P）

Register Name: PGSCALE Register address: 22(0x16) Read and write direction: R/W Default: 0x2710		
Bit	NAME	FUNCTION
15:0	PGSCALE[15:0]	Range：0~20000 This parameter is written in the factory using a high-precision turntable, please do not modify it
There is an error in the measurement of the Z-axis gyroscope, and a high-precision turntable will be used to measure this error when leaving the factory, and this parameter will be written into the sensor. Do not modify this parameter unless necessary. This parameter can be automatically calculated in the "scale factor mode". After entering the "scale factor mode", rotate the sensor by the angle set by "GSCALERANGE" to calculate the calibration factor. Z-axis calibration factorP=PGSCALE/10000.0 Example：		

Send：FF AA 27 16 00 （Read Z-axis calibration factorP） Return：55 5F 74 27 XX XX XX XX XX XX SUM 0x2774 = 10100，Read Z-axis calibration factorP = 10100/10000=1.01		

GSCALERANGE（Z-axis calibration angle）

Register Name: GSCALERANGE Register address: 24(0x18) Read and write direction: R/W Default: 0x02D0		
Bit	NAME	FUNCTION
15:0	GSCALERANGE[15:0]	When calibrating the factor, it needs to be calibrated according to this parameter
Example： Send：FF AA 18 68 01 （(Set the Z-axis calibration angle360）, 0x0168=360） The required rotation angle of the Z axis in the "scale factor mode" is generally set to an integer multiple of 360°.		

IICADDR（Device address）

Register Name: IICADDR Register Address: 26 (0x1A) Read and write direction: R/W Default: 0x0050		
Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address for I2C and Modbus communication 0x01~0x7F
示例：FF AA 1A 02 00（设置设备地址为0x02）		

LEDOFF（turn off LED light）

Register Name: LEDOFF Register Address: 27 (0x1B) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	LEDOFF	1: turn off LED light 0: turn on LED light
Example：FF AA 1B 01 00（turn off LED light）		

MAGRANGX~MAGRANGZ（Magnetic Field Calibration Range）

Register Name: MAGRANGX~MAGRANGZ Register Address: 28~30 (0x1C~0x1E) Read and write direction: R/W Default: 0x01F4		
Bit	NAME	FUNCTION
15:0	MAGRANGX[15:0]	Magnetic field calibration X-axis range
15:0	MAGRANGY[15:0]	Magnetic Field Calibration Y-axis Range
15:0	MAGRANGZ[15:0]	Magnetic field calibration Z-axis range
Example：FF AA 1C F4 01（Set the magnetic field calibration X-axis range to 500）		

BANDWIDTH

Register Name: BANDWIDTH Register Address: 31 (0x1F) Read and write direction: R/W Default: 0x0004		
Bit	NAME	FUNCTION
15:4		
3:0	BANDWIDTH[3:0]	set bandwidth 0000(0x00): 256Hz 0001(0x01): 188Hz 0010(0x02): 98Hz 0011(0x03): 42Hz 0100(0x04): 20Hz 0101(0x05): 10Hz 0110(0x06): 5Hz
Example：FF AA 1F 01 00 （set bandwidth to188Hz）		

GYRORANGE

Register Name: GYRORANGE Register Address: 32 (0x20) Read and write direction: R/W Default: 0x0003		
Bit	NAME	FUNCTION
15:4		
3:0	GYRORANGE[3:0]	Set the gyro range 0011(0x03): 2000°/s The default is 2000°/s, fixed and cannot be set
Example：FF AA 20 03 00（Set the gyro range 2000°/s）		

ACCRANGE（Accelerometer range）

Register Name: ACCRANGE Register Address: 33 (0x21) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:4		
3:0	ACCRANGE[3:0]	set accelerometer range 0000(0x00): ±2g 0011(0x03): ±16g This parameter cannot be set. The internal adaptive acceleration range of the product will automatically switch to 16g when the acceleration exceeds 2g.
Example：FF AA 21 00 00（Set the accelerometer range16g）		

SLEEP

Register Name: SLEEP Register Address: 34 (0x22) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	SLEEP	set sleep 1(0x01): sleep Any serial port data, can wake up
Example：FF AA 22 01 00（go to sleep）		

ORIENT

Register Name: ORIENT Register Address: 35 (0x23) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	ORIENT	Set the installation direction 0 (0x00): horizontal installation 1(0x01): vertical installation (the Y-axis arrow of the coordinate axis must be upward)
Example：FF AA 23 01 00（Set up vertical installation）		

AXIS6

Register Name: AXIS6 Register Address: 36 (0x24) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:1		
0	AXIS6	set algorithm 0(0x00): 9-axis algorithm (magnetic field solution navigation angle, absolute heading angle) 1(0x01): 6-axis algorithm (integral solution navigation angle, relative heading angle)
Example: FF AA 24 01 00 (set 6-axis algorithm mode)		

FILTK

Register Name: FILTK Register Address: 37 (0x25) Read and write direction: R/W Default: 0x001E		
Bit	NAME	FUNCTION
15:0	FILTK[15:0]	range：1~10000，The default is 30 (it is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 30) The smaller the FILTK[15:0], the stronger the seismic performance and the weaker the real-time performance. The larger the FILTK[15:0], the weaker the seismic performance and the stronger the real-time performance.
Example：FF AA 25 1E 00（Set the K value filter to 30）		

READADDR（read register）

Register Name: READADDR Register Address: 39 (0x27) Read and write direction: R/W Default: 0x00FF		
Bit	NAME	FUNCTION
15:8		
7:0	READADDR[7:0]	Read register range: Please refer to "Register Table"
Example: Send: FF AA 27 34 00 (read acceleration X axis 0x34) Return: 55 5F AXL AXH AYL AYH AZL AZH GXL GXH SUM For details, please refer to "Read Register Return Value" in the "Read Format" chapter		

ACCFILT（acceleration filter）

Register Name: ACCFILT Register Address: 42 (0x2A) Read and write direction: R/W Default: 0x01F4		
Bit	NAME	FUNCTION
15:0	ACCFILT[15:0]	Range: 1~10000, the default is 500 (it is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 500) The smaller the ACCFILT[15:0], the stronger the seismic performance and the weaker the real-time performance. The larger the ACCFILT[15:0], the weaker the seismic performance and the stronger the real-time performance. This parameter is an empirical value, which needs to be debugged according to different environments. In the tractor environment,

		ACCFILT[15:0] can be adjusted to 100, because the vibration of the tractor is serious and the anti-vibration performance needs to be improved
Example: Send: FF AA 2A F4 01 (set acceleration filter 500)		

POWONSEND

Register Name: POWONSEND Register Address: 45 (0x2D) Read and write direction: R/W Default: 0x0001		
Bit	NAME	FUNCTION
15:4		
3:0	POWONSEND[3:0]	Set the command to start: 0000(0x00): Turn off power-on data output 0001(0x01): Turn on power-on data output
Example: Send: FF AA 2D 00 00 (turn on power-on data output)		

VERSION

Register Name: VERSION Register Address: 46 (0x2E) Read and write direction: R Default: none		
Bit	NAME	FUNCTION
15:0	VERSION[15:0]	Different products, different version numbers
Example: FF AA 27 2E 00 (read version number, 0x27 means read, 0x2E is version number register) Return: 55 5F VL VH XX XX XX XX XX XX SUM VERSION[15:0]=(short)(((short)VH<<8)|VL)		

YYMM~MS（on-chip time）

Register Name: YYMM~MS Register address: 48~51 (0x30~0x33) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:8	YYMM[15:8]	mouth
7:0	YYMM[7:0]	year
15:8	DDHH[15:8]	hour
7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	second
7:0	MMSS[7:0]	minute
15:0	MS[15:0]	millisecond
Example： FF AA 30 16 03（Set year and month22-03） FF AA 31 0C 09（set day and hour12-09） FF AA 32 1E 3A（set minute and second30:58） FF AA 33 F4 01（set millisecond 500） Example： Send：FF AA 27 30 00（Read date, 0x27 means read, 0x30 is year month register）		

Return：55 5F YYMM[7:0] YYMM[15:8] DDHH[7:0] DDHH[15:8] MMSS[7:0] MMSS[15:8] MS[7:0] MS[15:8] SUM		

AX~AZ（acceleration）

Register Name: AX~AZ Register address: 52~54 (0x34~0x36) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X=AX[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s2)
15:0	AY[15:0]	Acceleration Y=AY[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s2)
15:0	AZ[15:0]	Acceleration Z=AZ[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s2)

GX~GZ（Angular velocity）

Register Name: GX~GZ Register address: 55~57 (0x37~0x39) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	GX[15:0]	Angular velocity X=GX[15:0]/32768*2000 °/s
15:0	GY[15:0]	Angular velocity Y=GY[15:0]/32768*2000 °/s
15:0	GZ[15:0]	Angular velocity Z=GZ[15:0]/32768*2000 °/s

HX~HZ（Magnetic field）

Register name: HX~HZ Register Address: 58~60 (0x3A~0x3C) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	HX[15:0]	Magnetic field X=HX[15:0] (unit: LSB)
15:0	HY[15:0]	Magnetic field Y=HY[15:0] (unit: LSB)
15:0	HZ[15:0]	Magnetic field Z=HZ[15:0] (unit: LSB)

Roll（Angle）

<table>
<tr><td>Register Name: Roll
Register address: 61~62 (0x3D~0x3E)
Read and write direction: R
Default: 0x0000</td><td></td><td></td></tr>
<tr><td>Bit</td><td>NAME</td><td>FUNCTION</td></tr>
<tr><td>15:0</td><td>LRoll[15:0]</td><td>Roll Angle X Low</td></tr>
<tr><td>15:0</td><td>HRoll[15:0]</td><td>Roll angle X high position, roll angle X=(((int)HRoll<<16)| LRoll)/1000°</td></tr>
</table>

Pitch（Angle）

Register Name: Pitch Register address: 63~64 (0x3F~0x40) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	LPitch[15:0]	Pitch angle Y low
15:0	HPitch[15:0]	Pitch angle Y high position, pitch angle X= (((int)HPitch<<16)| LPitch)/1000°

Yaw（Angle）

Register Name: Yaw Register address: 65~66 (0x41~0x42) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	LYaw[15:0]	Heading Angle Z Low
15:0	HYaw[15:0]	Heading angle Z high position, heading angle Z=(((int)HYaw<<16)| LYaw)/1000°

TEMP（Angle）

Register Name: TEMP Register Address: 67 (0x43) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	TEMP[15:0]	temperature=TEMP[15:0]/ 100℃

PressureL~HeightH（barometric altitude）

<table>
<tr><td>Register Name: PressureL~HeightH
Register address: 69~72 (0x45~0x48)
Read and write direction: R
Default: 0x0000</td><td></td><td></td></tr>
<tr><td>Bit</td><td>NAME</td><td>FUNCTION</td></tr>
<tr><td>15:0</td><td>PressureL[15:0]</td><td>barometric = ((int)PressureH[15:0] <<16)|PressureL[15:0] (Pa)</td></tr>
<tr><td>15:0</td><td>PressureH[15:0]</td><td></td></tr>
<tr><td>15:0</td><td>HeightL[15:0]</td><td>altitude= ((int)HeightH[15:0] <<16)|HeightL[15:0](cm)</td></tr>
<tr><td>15:0</td><td>HeightH[15:0]</td><td></td></tr>
</table>

q0~q3（Quaternion）

Register name: q0~q3 Register address: 81~84 (0x51~0x54) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	q0[15:0]	quaternion0=q0[15:0]/32768
15:0	q1[15:0]	quaternion1=q1[15:0]/32768
15:0	q2[15:0]	quaternion2=q2[15:0]/32768
15:0	q3[15:0]	quaternion3=q3[15:0]/32768

GYROCALITHR（Gyro Still Threshold）

Register Name: GYROCALITHR Register Address: 97 (0x61) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	GYROCALITHR[15:0]	Set the gyroscope inactivity threshold: Gyro gyroscope static threshold=GYROCALITHR [15:0]/1000(°/s)
Example: FF AA 61 32 00 (set the gyro static threshold to 0.05°/s, 0x0032=50, 50/1000=0.05(°/s)) When the angular velocity change is less than 0.05°/s and lasts for the time of "GYROCALTIME", the sensor recognizes it as stationary and automatically resets the angular velocity less than 0.05°/s to zero		

The setting rule of the static threshold of the gyroscope can be determined by reading the value of the "WERROR" register. The general setting rule is: GYROCALITHR=WERROR *1.2, unit: °/s This register needs to be used in conjunction with the GYROCALTIME register		

GYROCALTIME（Gyro auto calibration time）

Register Name: GYROCALTIME Register Address: 99 (0x63) Read and write direction: R/W Default: 0x03E8		
Bit	NAME	FUNCTION
15:0	GYROCALTIME[15:0]	Set gyroscope auto-calibration time
Example: Set gyroscope auto-calibration time to 500ms FF AA 63 F4 01 (set gyroscope automatic calibration time 500ms, 0x01F4=500(ms)) When the angular velocity change is less than "GYROCALITHR" and lasts for 500ms, the sensor recognizes that it is stationary and automatically resets the angular velocity less than 0.05°/s to zero This register needs to be used in conjunction with the GYROCALITHR register		

KEY（unlock）

Register Name: KEY Register Address: 105 (0x69) Read and write direction: R/W Default: 0x0000		
Bit	NAME	FUNCTION
15:0	KEY[15:0]	Unlock register: When performing a write operation, you need to set this register first
Example: Unlock, write 0xB588 to this register (other values are invalid) Send: FF AA 69 88 B5		

WERROR（Gyroscope change value）

Register Name: WERROR Register Address: 106 (0x6A) Read and write direction: R Default: 0x0000		
Bit	NAME	FUNCTION
15:0	WERROR[15:0]	Gyroscope change value=WERROR[15:0]/1000*180/3.1415926(°/s) When the sensor is stationary, the "GYROCALITHR" register can be set by changing this register

WZTIME（Angular velocity continuous rest time）

Register Name: WZTIME Register Address: 110 (0x6E) Read and write direction: R/W Default: 0x01F4		
Bit	NAME	FUNCTION
15:0	WZTIME[15:0]	Angular velocity continuous rest time
Example： FF AA 6E F4 01（Set the angular velocity continuous still time500ms，0x01F4=500(ms)） When the angular velocity is less than "WZSTATIC" and lasts for 500ms, the angular velocity output is 0, and the Z-axis heading angle is not integrated This register needs to be used in conjunction with the "WZSTATIC" register		

WZSTATIC（Angular velocity integral threshold）

Register Name: WZSTATIC Register Address: 111 (0x6F) Read and write direction: R/W Default: 0x012C		
Bit	NAME	FUNCTION
15:0	WZSTATIC[15:0]	Angular velocity integral threshold=WZSTATIC[15:0]/1000(°/s)
Example: FF AA 6F F4 01 (set the angular velocity integral threshold to 0.5°/s, 0x01F4=500, 500/1000=0.5(°/s)) When the angular velocity is greater than 0.5°/s, the Z-axis heading angle starts to integrate the acceleration When the angular velocity is less than 0.5°/s, and the duration set by the register "WZTIME", the angular velocity output is 0, and the Z-axis heading angle is not integrated This register needs to be used in conjunction with the "WZTIME" register		

NUMBERID1~NUMBERID6（Device number）

Register Name: NUMBERID1~NUMBERID6 Register address: 127~132 (0x7F~0x84) Read and write direction: R Default: none		
Bit	NAME	FUNCTION
15:0	NUMBERID1[15:0]	
15:0	NUMBERID2[15:0]	
15:0	NUMBERID3[15:0]	
15:0	NUMBERID4[15:0]	
15:0	NUMBERID5[15:0]	
15:0	NUMBERID6[15:0]	
Device number：WT4200000001		

XREFROLL~YREFPITCH（Angle zero reference value）

Register Name: NUMBERID1~NUMBERID6 Register address: 127~132 (0x7F~0x84) Read and write direction: R/W Default: none		
Bit	NAME	FUNCTION
15:0	LREFROLL[15:0]	Roll angle zero reference value low
15:0	HREFROLL[15:0]	Roll angle zero reference value high, roll angle zero reference= (((int)HREFROLL<<16)|LREFROLL)/1000(°)
15:0	LREFPITCH[15:0]	Pitch angle zero reference value low
15:0	HREFPITCH[15:0]	Pitch angle zero reference value= (((int)HREFPITCH<<16)|LREFPITCH)/1000(°)
Example: The current roll angle is 2°, set the roll angle zero, subtract 2°, then REFROLL[31:0]=2*1000=2000=0x07D0 LREFROLL[15:0]=0x07D0, HREFROLL[15:0]=0; FF AA 95 D0 07 (set "LREFROLL")		

FF AA 96 00 00 (set "HREFROLL")		

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